

Miscellaneous Algebra Exercises

Polynomials

1. Show that $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$.
2. Show that $\sqrt{n} = 2^{n-1} \prod_{k=1}^{n-1} \sin \frac{k\pi}{2n}$.
3. The proposition: for any real numbers $a_1, a_2, \dots, a_n$ ($n > 2$) the following inequality holds:

$$\begin{aligned} & (a_1 - a_2)(a_1 - a_3) \dots (a_1 - a_n) + \\ & (a_2 - a_1)(a_2 - a_3) \dots (a_2 - a_n) + \\ & \\ & (a_n - a_1)(a_n - a_2) \dots (a_n - a_{n-1}) \geq 0 \end{aligned}$$

is true for $n = 3, 5$ only.

4. A number is *good* if the sum of its digits is 18, (for example 130518 is a good number), find the number of good 6-digit numbers.
5. How many odd coefficients are there in the expansion of $(x^2 - x + 1)^{2009}$?
6. Denote the number of odd coefficients of a polynomial $p(x)$ by $w(p)$. Let $q_i(x) = (1+x)^{i_i}$. Let $i_1, i_2, \dots, i_n$ be integers satisfying $0 \leq i_1 < i_2 < \dots < i_n$. show that $w(q_{i_1} + q_{i_2} + \dots + q_{i_n}) \geq w(q_{i_1})$.
6. Let p be an odd prime number. Find the number of subsets A of the set $\{1, 2, \dots, 2p\}$ such that
 - (a) A has exactly p elements, and
 - (b) the sum of all the elements in A is divisible by p .
7. Factorize $(a+b+c)^3 - (a^3 + b^3 + c^3)$.
8. $(1+x^n+x^{2n}+\dots+x^{mn})$ divisible by $(1+x+x^2+\dots+x^m)$.
9. $n > 1$, can $f(x) = x^n - 5x^{n-1} + 3$ be factored into two non-trivial polynomials with

integer coefficients?

10. Find all polynomials f such that f has non-negative integer coefficients, $f(1) = 7$ and $f(2) = 2017$.

11. Determine the maximum value of $m^2 + n^2$, where m, n are integers satisfying $1 \leq m, n \leq 1981$ and $(n^2 - mn - m^2)^2 = 1$.

12. Let a, b be real numbers. It is given that the equation $x^4 + ax^3 + bx^2 + ax + 1 = 0$ has at least one real root. Find the minimum value of $a^2 + b^2$.

13. The roots of $x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$ cannot be *all* real if $2a^2 < 5b$.

14. The product of two of the four roots of $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ is -32 . Find k .

15. If a and b are two roots of $x^4 + x^3 - 1 = 0$, show that ab is a root of $x^6 + x^4 + x^3 - x^2 - 1 = 0$.

16. a, b, c distinct integers, no polynomial with integer coefficients and such that $f(a) = b, f(b) = c, f(c) = a$.

17. The sequence of polynomials $\{P_n(x)\}$ is defined as follows: $P_1(x) = x^2 - 2$ and $P_n(x) = P_1(P_{n-1}(x))$. Show that for all positive n the roots $P_n(x) = x$ are distinct real numbers.

18. Least number of linear factors needed to be removed from the equation

$$(x-1)(x-2)\dots(x-2016) = (x-1)(x-2)\dots(x-2016)$$

so that the resulting equation has no real root.

19. Find all real polynomials $P(x)$ which satisfy the equality $P(a-b) + P(b-c) + P(c-a) = 2P(a+b+c)$ for all a, b, c satisfying $ab + bc + ca = 0$.

20. Given positive integers a and b and such that $ab+1$ divides a^2+b^2 , then $\frac{a^2+b^2}{ab+1}$ is a perfect square.

21. $p(x)$ a polynomial with integer coefficients and such that both $p(0)$ and $p(1)$ are odd. Then $p(x)$ has no integer solutions.

22. $f(x) = (x-a)(x-b)(x-c)(x-d) - 9$, where a, b, c, d are distinct integers. Suppose $f(x)$ has an integer solution. Show that $a+b+c+d$ is divisible by 4.

23. $f(x) = a_0 + a_1x + a_2x^2 + \dots + a_nx^n$ and such that all its roots are real, also $a_i = \pm 1$.

Find all possible polynomials.

24. $p(x)$ an integer polynomial and let $n(p)$ the number of integers k such that $p(k)^2 = 1$. Show that $n(p) \leq \deg p + 2$.

25. Find all roots, real and complex, of the system

$$\begin{aligned}x + y + z &= 3, \\x^2 + y^2 + z^2 &= 3, \\x^3 + y^3 + z^3 &= 3.\end{aligned}$$

26. Given

$$\begin{aligned}x + y + z &= 3, \\x^2 + y^2 + z^2 &= 5, \\x^3 + y^3 + z^3 &= 12.\end{aligned}$$

Find $x^4 + y^4 + z^4$.

27. Let $f(x)$ be a polynomial with real coefficients and such that $f(x) \geq 0$ for all real x . Show that $f(x)$ is the sum of squares of two real polynomials.

28. $5^{1985} - 1$ is the product of three factors, each larger than 5^{100} .

29. $x^4 + x^3 + x^2 + x + 1$ is the difference of the squares of two real polynomials (with different degrees) and there are four such solutions.

30. $Q(t)$ a real polynomial with two distinct real roots. Show that there exists a non-constant polynomial with leading coefficient 1 such that the absolute values of all coefficients of $Q(P(x))$ other than the leading one are less than 0.001.

31. Does there exist a polynomial $P(x)$ with integer coefficients and such that

$$P(1 + \sqrt[3]{2}) = 1 + \sqrt[3]{2} \quad \text{and} \quad P(1 + \sqrt{5}) = 2 + 3\sqrt{5}?$$

32. $r(x)$ a real polynomial of odd degree, there exist only finitely many pairs of real polynomials $p(x)$ and $q(x)$ such that $p(x)^3 + q(x)^2 = r(x)$.

33. Find all polynomials $P(x, y)$ such that (i) P is homogeneous of degree n , i.e. $P(tx, ty) = t^n P(x, y)$, (ii) for all a, b, c , $P(a+b, c) + P(b+c, a) + P(c+a, b) = 0$, (i) $P(1, 0) = 1$.

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Evaluate

Show that $\cos \frac{\pi}{7} - \cos \frac{2\pi}{7} + \cos \frac{3\pi}{7} = \frac{1}{2}$.

(IMO1963) Let $z = \cos \frac{\pi}{7} + i \sin \frac{\pi}{7}$. Then $z^7 + 1 = 0$. Factorize $z^7 + 1$ to get

$z^6 - z^5 + z^4 - z^3 + z^2 - z + 1 = 0$. Hence $z^3 - z^2 + z = \frac{1}{1 - z^3}$. Thus we have to find the

real part of $z^3 - z^2 + z$. Finally we have to prove for any complex t of norm 1, the

real part of $\frac{1}{1-t}$ is $\frac{1}{2}$. Now $\frac{1}{1-t} = \frac{1}{1-e^{i\theta}} = \frac{1}{1-e^{i\theta}} \cdot \frac{1-e^{-i\theta}}{1-e^{-i\theta}} = \frac{(1-\cos\theta) + i\sin\theta}{2-2\cos\theta},$

and its real part is $\frac{1}{2}$.

Show that $\sqrt{n} = 2^{n-1} \prod_{k=1}^{n-1} \sin \frac{k\pi}{2n}$.

(Unknown) Consider the $2n$ roots of $f(x) = x^{2n} - 1$, which are $e^{i \frac{2k\pi}{2n}}, i = 0, 1, \dots, 2n-1$. We get

$$f(x) = (x-1)(x+1) \prod_{k=1}^{n-1} (x - e^{i\frac{2k\pi}{2n}}) \prod_{k=1}^{n-1} (x - e^{i\frac{(2n-k)\pi}{2n}})$$

$$= (x^2 - 1) \prod_{k=1}^{n-1} (x^2 - 2x \cos \frac{2k\pi}{2n} + 1).$$

Hence $x^{2^{n-2}} + x^{2^{n-4}} + \dots + x^2 + 1 = \prod_{k=1}^{n-1} (x^2 - 2x \cos \frac{2k\pi}{2n} + 1)$.

Put $x = 1$, get

$$n = \prod_{k=1}^{n-1} (2 - 2 \cos \frac{2k\pi}{2n}) = \prod_{k=1}^{n-1} 2^2 \sin^2 \frac{k\pi}{2n} = 2^{2n-2} \prod_{k=1}^{n-1} \sin^2 \frac{k\pi}{2n},$$

and the result.

The proposition: for any real numbers $a_1, a_2, \dots, a_n$ ($n > 2$) the following inequality holds:

$$\begin{aligned} & (a_1 - a_2)(a_1 - a_3) \dots (a_1 - a_n) + \\ & (a_2 - a_1)(a_2 - a_3) \dots (a_2 - a_n) + \\ & \\ & (a_n - a_1)(a_n - a_2) \dots (a_n - a_{n-1}) \geq 0 \end{aligned}$$

Is true for $n = 3, 5$ only.

(IMO1971) Consider $p(x) = (x - a_1)(x - a_2) \dots (x - a_n)$, the required sum is $S_n(a_1, a_2, \dots, a_n) = \sum_{j=1}^n p'(a_j)$. The required sum is symmetric, hence may assume $a_1 > a_2 > \dots > a_n$. Then prove the inequality true for $n = 3, 5$, false for $n = 4$. For $n > 5$, consider $a_1 = a_2 = \dots = a_{n-4} = 0$, $a_{n-3} = a_{n-2} = a_{n-1} = 2$ and $a_n = 1$. Get $p(x) = x^{n-4}(x-2)^3(x-1)$, take derivatives to get the result.

Count

A number is *good* if the sum of its digits is 18, (for example 130518 is a good number), find the number of good 6-digit numbers.

(August Test 2013) Using generating functions, the number of good 6-digit numbers equals to the coefficient of z^{18} in the expansion of

$$A(z) = (z + z^2 + \dots + z^9)(1 + z + z^2 + \dots + z^9)^5. \text{ Now } z + z^2 + \dots + z^9 = \frac{z(1-z^9)}{1-z} \text{ and}$$

$$(1 + z + z^2 + \dots + z^9)^5 = \left(\frac{1-z^{10}}{1-z}\right)^5. \text{ Hence } A(z) = z(1-z^9)(1-z^{10})^5(1-z)^{-6}. \text{ But}$$

$$z(1-z^9)(1-z^{10})^5 = z(1-z^9)(1 - C_1^5 z^{10} + \dots) = z - z^{10} - 5z^{11} + \dots \text{ and}$$

$$(1-z)^{-6} = 1 + C_1^6 z + C_2^7 z^2 + C_3^8 z^3 + \dots.$$

Therefore the coefficient of z^{18} in the expansion of $A(z)$ is $= C_{17}^{22} - C_8^{13} - 5C_7^{12} = 21087$, which equals the number of good numbers. (Remark: the problem may be modeled as placing 18 balls (indistinguishable) into 5 boxes (distinguishable) with the first box has 1 ball or more, but no box may have 10 balls or more.)

How many odd coefficients are there in the expansion of $(x^2 - x + 1)^{2009}$?

(October Test 2009) Since we are interested only in the parity of the coefficients, we take modulo 2 on the coefficients throughout (so every coefficient is either 0 or 1). Let $f(n)$ be the number of odd coefficients in the expansion of $(x^2 - x + 1)^n$. To find $f(2009)$, we need a series of observations.

Observation 1. $f(2n) = f(n)$.

This follows from the fact that for pairwise distinct $a_1, a_2, \dots, a_n$, we have

$$(x^{a_1} + x^{a_2} + \cdots + x^{a_n})^2 = x^{2a_1} + x^{2a_2} + \cdots + x^{2a_n} + 2 \sum_{i \neq j} x^{a_i} x^{a_j} \equiv x^{2a_1} + x^{2a_2} + \cdots + x^{2a_n}.$$

Observation 2. $f(2^k) = 3$ for all non-negative integers k .

This follows from the fact that $(x^2 - x + 1)^2 \equiv (x^2 + x + 1)^2 = x^4 + 2x^3 + 3x^2 + 2x + 1 \equiv x^4 + x^2 + 1$, so we have $(x^2 - x + 1)^{2^k} \equiv x^{2^{k+1}} + x^{2^k} + 1$ for all non-negative integers k .

Observation 3. $f(2^k a + b) = f(a)f(b)$ for positive integers a, b where $b < 2^{k-1}$.

As remarked in Observation 2, we have $(x^2 - x + 1)^{2^k a} \equiv (y^2 + y + 1)^a$ where $y = x^{2^k}$.

On the other hand, $(x^2 - x + 1)^b$ is of degree $2b < 2^k$. We thus get the desired conclusion.

Observation 4. $f(2^k - 1) = \frac{2^{k+2} - (-1)^k}{3}$ for all positive integers k .

Observe the coefficients of $(x^2 - x + 1)^{2^k - 1}$ for $k = 1, 2, 3, \dots$, we see the pattern as follows: 111, 1101011, $(110)^2(111)(011)^2$ (this means

$(x^2 - x + 1)^{2^3 - 1} \equiv (1x^0 + 1x^1 + 0x^2) + (1x^3 + 1x^4 + 0x^5) + (1x^6 + 1x^7 + 1x^8) + (0x^9 + 1x^{10} + 1x^{11}) + (0x^{12} + 1x^{13} + 1x^{14})$, same for below), $(110)^5(1)(011)^5$, $(110)^{10}(111)(011)^{10}$, ..., and we can prove by induction that such pattern (i.e. $(110)^{\frac{2^k-2}{3}}(111)(011)^{\frac{2^k-2}{3}}$ for odd k and $(110)^{\frac{2^k-1}{3}}(1)(011)^{\frac{2^k-1}{3}}$ for even k) does indeed carry on.

With the above observations, we can proceed to find $f(2009)$. Since 2009 is 11111011001 in base 2, we have $2009 = 2^5(2^5 - 1) + 2^3(2^2 - 1) + 1$. It follows from the above observations that

$$f(2009) = f(2^5 - 1)f(2^2 - 1)f(1) = \frac{2^7 + 1}{3} \times \frac{2^4 - 1}{3} \times 3 = 645.$$

Remark. This is a difficult question. While an apparently similar question appeared in last year's Prelim (on binomial expansion), this question (on trinomial expansion) involves considerably more techniques. Considering the binary notation of 2009 should be the natural approach (especially if one works out the first few powers and finds that $f(2^k) = 3$ for all k), and the most difficult part should be to find $f(2^k - 1)$ in Observation 4, as it takes effort to compute enough data to look for patterns.

Denote the number of odd coefficients of a polynomial $p(x)$ by $w(p)$. Let

$q_i(x) = (1+x)^i$. Let $i_1, i_2, \dots, i_n$ be integers satisfying $0 \leq i_1 < i_2 < \dots < i_n$. show that

$$w(q_{i_1} + q_{i_2} + \dots + q_{i_n}) \geq w(q_{i_1}).$$

(IMO1985) delicate inductions: suppose it is true for all sequences $0 \leq i_1 < i_2 < \dots < i_n$

where $i_n < 2^m$ and then it is true for all sequences $0 \leq i_1 < i_2 < \dots < i_n$ where

$$2^m \leq i_n < 2^{m+1}.$$

Let p be an odd prime number. Find the number of subsets A of the set $\{1, 2, \dots, 2p\}$

such that

(a) A has exactly p elements, and

(b) the sum of all the elements in A is divisible by p .

(IMO1995.6) Let $\xi = e^{2\pi i/p}$ be a primitive root of unity. The idea is to consider products of p elements within the list $\xi^1, \xi^2, \dots, \xi^{2p}$.

Then

$$\prod_{i=1}^{2p} (x - \xi^i) = (x^p - 1)^2 = x^{2p} - 2x^p + 1.$$

By Viète's formula applied to the coefficient of x^p , get

$$2 = \sum_{1 \leq i_1 \leq \dots \leq i_p \leq 2p} \xi^{i_1 + \dots + i_p} = \sum_{j=0}^{p-1} n_j \xi^j,$$

Where n_j is the number of subsets $\{i_1, i_2, \dots, i_p\}$ of $\{1, 2, \dots, 2p\}$ such that

$$i_1 + i_2 + \dots + i_p = j \pmod{p}.$$

We look for the number n_0 .

But ξ is also a root of the polynomial of degree $p-1$, namely

$$P(x) = n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j.$$

On the other hand, the minimal polynomial of ξ is the cyclotomic polynomial

$$\phi_p(x) = 1 + x + x^2 + \dots + x^{p-1}.$$

It follows that they differ by a constant c ,

$$n_0 - 2 + \sum_{j=1}^{p-1} n_j x^j = c(1 + x + x^2 + \dots + x^{p-1}).$$

By setting $x = 1$, we find

$$c = \frac{1}{p} \left(\sum_{j=0}^{p-1} n_j - 2 \right) = \frac{1}{p} \left(\binom{2p}{p} - 2 \right).$$

(Consider how all the n_j 's selected from 1 to $2p$ to get $\binom{2p}{p}$).

Using c , we obtain the value of n_0 by comparing the constant terms:

$$n_0 = 2 + \frac{1}{p} \left(\binom{2p}{p} - 2 \right).$$

Reducibility

Factorize $(a+b+c)^3 - (a^3 + b^3 + c^3)$.

(Unknown) Consider $P(a,b,c) = (a+b+c)^3 - (a^3 + b^3 + c^3)$ as a polynomial of b . When $b = -a$, get $P(a, -a, c) = 0$, thus $(b+a)$ is a factor. Likewise $(c+b)$ and $(a+c)$, hence $(a+b+c)^3 - (a^3 + b^3 + c^3) = k(a+b)(b+c)(c+a)$. Put $a = b = c = 1$, get $K = 3$.

$(1 + x^n + x^{2n} + \dots + x^{mn})$ divisible by $(1 + x + x^2 + \dots + x^m)$.

(USAMO1977) Same as $\frac{x^{n(m+1)} - 1}{x^n - 1}$ divisible by $\frac{x^{m+1} - 1}{x - 1}$, namely $\frac{x^{n(m+1)} - 1}{x^{m+1} - 1} \frac{x - 1}{x^n - 1}$ a polynomial. As all factors of $x^{n(m+1)} - 1$ are distinct, $x^{m+1} - 1$ and $x^n - 1$ has no common factor other than $x - 1$, thus $m+1$ and n must be relatively prime, condition also sufficient since $x^{n(m+1)} - 1 = (x^n)^{m+1} - 1 = (x^{m+1})^n - 1$ and is divisible by both $x^n - 1$ and $x^{m+1} - 1$.

$n > 1$, can $f(x) = x^n - 5x^{n-1} + 3$ be factored into two non-trivial polynomials with integer coefficients?

(IMO1993) Note ± 3 are not roots of the polynomial, hence if it is decomposable, it is of the form $(x^r + a_{r-1}x^{r-1} + \dots + a_1x + a_0)(x^s + b_{s-1}x^{s-1} + \dots + b_1x + b_0)$, with

$r > 1, s > 1$ and $r + s = n$. Since $3 = a_0 b_0$, we may assume $3 \mid a_0$ but $3 \nmid b_0$. When $r \leq s$, get

$$\begin{aligned} a_0 b_1 + a_1 b_0 &= 0 \\ a_0 b_2 + a_1 b_1 + a_2 b_0 &= 0 \\ &\dots\dots\dots \\ a_0 b_{r-1} + \dots + a_{r-2} b_1 + a_{r-1} b_0 &= 0 \\ a_0 + \dots + a_{r-1} b_1 + b_0 &= 0. \end{aligned}$$

Successively we get 3 divides $a_0, a_1, \dots, a_{r-1}, b_0$, contradiction. Similar consideration when $r > s$, we get $3 \mid a_1, \dots, 3 \mid a_s$. Let i be the greatest index, $i \geq s$ such that

$3 \mid a_i$. Such an index exists, since 3 does not divide a_r . Then 3 does not divide a_{i+1} and $s < i+1 \leq r \leq n-2$. The coefficient of the monomial x^{i+1} in $f(x)$ is

$$0 = b_0 a_{i+1} + b_1 a_i + \dots + b_{s-1} a_{i-s} + b_s a_{i-s+1}.$$

Since $3 \mid a_j$ for all $j < i+1$ and 3 does not divide $b_0 a_{i+1}$, we obtain a contradiction.

Values, Roots and Coefficients

Find all polynomials f such that f has non-negative integer coefficients, $f(1) = 7$ and $f(2) = 2017$.

(August Test 2017) Indeed there is only one such polynomial, $f(x) = x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + 1$.

Let $g(x)$ be a polynomial with non-negative integer coefficients and such that $g(2) = 2017$. Suppose some coefficient of $g(x)$ is at least 2, say the coefficient

x^k . Define $h(x) = g(x) - 2x^k + x^{k+1}$. Then $h(2) = g(2)$ and $h(1) < g(1)$. Proceed

in this matter, we can find some polynomial $q(x)$ with coefficients 0 or 1, and such that

$q(1) < g(1)$. Now there is a unique polynomial with coefficients 0 or 1, and has value 2017 when $x = 2$, namely $x^{10} + x^9 + x^8 + x^7 + x^6 + x^5 + 1$, (binary expansion of 2017). This polynomial assumes value 7 when $x = 1$.

Determine the maximum value of $m^2 + n^2$, where m, n are integers satisfying $1 \leq m, n \leq 1981$ and $(n^2 - mn - m^2)^2 = 1$.

(IMO1981) If $m = n$, then $m = n = 1$, and $(1, 1)$ a solution. Also $(0, 1), (1, 0)$ are solutions. Note if $n \neq m$ and (n, m) a positive solution, then $n > m > n - m$ and $(m, n - m)$ a solution. By descent and ascent, we will eventually get to $(1, 1)$ and $(1, 0)$. Hence we get a list of numbers correspond to the Fibonacci numbers:
 $(1, 0) \rightarrow (1, 1) \rightarrow (2, 1) \rightarrow (3, 2) \rightarrow (5, 3) \rightarrow \dots$ Hence to obtain the maximum value of $m^2 + n^2$, we consider the sequence:
 $0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, 1597, \dots$ The required maximum is $987^2 + 1597^2$.

Let a, b be real numbers. It is given that the equation $x^4 + ax^3 + bx^2 + ax + 1 = 0$ has at least one real root. Find the minimum value of $a^2 + b^2$.

(August Test 2013) Let $y = x + \frac{1}{x}$. The given condition means that the equation $y^2 + ay + (b - 2) = 0$ has at least one root whose absolute value ≥ 2 . Hence $\Delta = a^2 - 4(b - 2) \geq 0$.

Further, $2^2 + 2a + (b - 2) \leq 0$ or $(-2)^2 + (-2)a + (b - 2) \leq 0$, which gives (i) $a^2 \geq 4b - 8$ and (ii) $2a + b + 2 \leq 0$ or $2a - b - 2 \geq 0$.

Draw the graphs of (i) $b \leq \frac{a^2}{4} + 2$ (ii) $b \leq -2a - 2$ (iii) $b \leq 2a - 2$.

The feasible region is the region below the line $b = -2a - 2$ or $b = 2a - 2$.

Consider the circle, centred at $(0, 0)$, that touches these two lines.

Since the distance from $(0, 0)$ to $2a + b + 2 = 0$ is $2/\sqrt{5}$, the required circle is $a^2 + b^2 = 4/5$.

Hence the minimum value of $a^2 + b^2$ is $4/5$.

(Remarks: Indeed the equation is reciprocal, hence if x is a root, so is $\frac{1}{x}$. Let

$y = x + \frac{1}{x}$, we get the equation $y^2 + ay + (b - 2) = 0$. We get y_1, y_2 be the two roots.

As the root(s) are real, we have the positive discriminant and hence $y \geq 2$ or $y \leq -2$.

Using Vieta, we get

$$a^2 + b^2 = y_1^2 y_2^2 + y_1^2 + y_2^2 + 6y_1 y_2 + 4.$$

If $y_1 \geq 2$, one shows that $a^2 + b^2 \geq 5(y_2 + \frac{6}{5})^2 + \frac{4}{5} \geq \frac{4}{5}$. Likewise

If $y_1 \leq -2$, one shows that $a^2 + b^2 \geq 5(y_2 - \frac{6}{5})^2 + \frac{4}{5} \geq \frac{4}{5}$.

In any case we obtain the desired maximum.)

The roots of $x^5 + ax^4 + bx^3 + cx^2 + dx + e = 0$ cannot be all real if $2a^2 < 5b$.

(USAMO1983) Let the roots be $r_i, 1 \leq i \leq 5$. Then $-a = \sum r_i$ and $b = \sum r_i r_j$. Then

$2a^2 < 5b$ is equivalent to $2\left\{\sum r_i\right\}^2 - 5\sum r_i r_j < 0$, or $\sum (r_i - r_j)^2 < 0$, impossible.

The product of two of the four roots of $x^4 - 18x^3 + kx^2 + 200x - 1984 = 0$ is -32 . Find k .

(USAMO1984) Let $r_i, 1 \leq i \leq 4$ be the four roots. If $r_1 r_2 = -32$, then

$$r_3 r_4 = \frac{r_1 r_2 r_3 r_4}{r_1 r_2} = \frac{-1984}{-32} = 62. \text{ Hence}$$

$$x^4 - 18x^3 + kx^2 + 200x - 1984 = (x - r_1)(x - r_2)(x - r_3)(x - r_4) = (x^2 - px - 32)(x^2 - qx + 62).$$

Equating the coefficients, get $p + q = 18, -62p + 32q = 200, k = 62 + pq - 32$. Solve for p and q to get $k = 86$.

If a and b are two roots of $x^4 + x^3 - 1 = 0$, show that ab is a root of $x^6 + x^4 + x^3 - x^2 - 1 = 0$.

(USAMO 1977) Let a, b, c, d be the four roots of the quartic equation. Let $p = a + b, q = ab, r = c + d, s = cd$. Then by comparing coefficients of the quartic equation, get

- (1) $-1 = a + b + c + d = p + r,$
- (2) $0 = ab + ac + ad + bc + bd + cd = pr + q + s,$
- (3) $0 = abc + bcd + cda + dab = ps + qr,$
- (4) $-1 = abcd = qs.$

From (1) and (4), get $r = -1 - p, s = -1/q$, put into (2), (3), get $p(1 + p) = q - \frac{1}{q}$

and $p = \frac{-q^2}{q^2 + 1}$. Finally eliminate p to get $q^6 + q^4 + q^3 - q^2 - 1 = 0$.

a, b, c distinct integers, no polynomial with integer coefficients and such that $f(a) = b, f(b) = c, f(c) = a$.

(USAMO 1974) Note $a - b$ divides $f(a) - f(b) = b - c$, and likewise, thus

$|a - b| = |b - c| = |c - a|$, impossible for distinct integers.

The sequence of polynomials $\{P_n(x)\}$ is defined as follows: $P_1(x) = x^2 - 2$ and $P_n(x) = P_1(P_{n-1}(x))$. Show that for all positive n the roots $P_n(x) = x$ are distinct real numbers.

(IMO1976) Note $P_1(x)$ transform both intervals $[-2, 0]$ and $[0, 2]$ onto $[-2, 2]$, and $[-2, 2] = [-2, 0] \cup [0, 2]$. Thus $P_2(x) - x = P_1(P_1(x)) - x$ again both intervals $[-2, 0]$ and $[0, 2]$ onto $[-2, 2]$, hence there are two distinct roots in $[-2, 0]$ and $[0, 2]$. Further we make induction on n and observe that $\deg(P_n(x) - x) = 2^n$. (This problem may also be solved by substituting $x = 2 \cos t$)

Least number of linear factors needed to be removed from the equation

$$(x-1)(x-2)\dots(x-2016) = (x-1)(x-2)\dots(x-2016)$$

so that the resulting equation has no real root.

(IMO2016) Clearly at least 2016 factors needed to be removed, and indeed it suffices. Namely remove the factors $(x-k)$ with $k \equiv 2, 3 \pmod{4}$ from the left, and the factors $(x-k)$, with $k \equiv 1, 4 \pmod{4}$ from the right.

Find all real polynomials $P(x)$ which satisfy the equality $P(a-b) + P(b-c) + P(c-a) = 2P(a+b+c)$ for all a, b, c satisfying $ab+bc+ca=0$.

(IMO2004) If $(a, b, c) = (6x, 3x, -2x)$, then $ab+bc+ca=0$. This triple gives

$P(3x) + P(5x) + P(-8x) = 2P(7x)$ for all x . Let $P(x) = \sum a_i x^i$, put into the equation,

get $(3^i + 5^i + (-8)^i - 2 \cdot 7^i) a_i = 0$. For odd i , the brackets are negative. For $i=0$, and even $i \geq 6$, the brackets are positive. For $i=2, 4$, the brackets are 0. Thus $P(x) = ax^2 + bx^4$.

Given positive integers a and b and such that $ab+1$ divides a^2+b^2 , then $\frac{a^2+b^2}{ab+1}$ is a perfect square.

(IMO1998) Let $\frac{a^2+b^2}{ab+1}=k$ and such that k is not a perfect square. After re-arranging we have $a^2-kab+b^2=k$, with $a>0$ and $b>0$. Assume now (a_0, b_0) is a solution of the Diophantine equation and such that a_0+b_0 is as *small* as possible. By symmetry we may assume $a_0 \geq b_0 > 0$. Fixing b_0 and k , we may assume a_0 is a solution of the quadratic equation $x^2-kb_0x+b_0^2-k=0$. Now let another root of the equation be a' . Using sum and product of roots, we have $a_0+a'=kb_0$ and $a_0a'=b_0^2-k$. The first equation implies a' is an integer. The second equation implies $a' \neq 0$, otherwise k is a perfect square, contradicting our hypothesis. Now a' also cannot be negative, otherwise $a'^2-ka'b_0+b_0^2 \geq a'^2+k+b_0^2 > k$. Hence $a' > 0$. Finally $a' = \frac{b_0^2-k}{a_0} \leq \frac{b_0^2-1}{a_0} \leq \frac{a_0^2-1}{a_0} < a_0$. This implies (a', b_0) is an integer solution of $a^2-kab+b^2=k$, and $a'+b_0 < a_0+b_0$, contradicting the minimality of a_0+b_0 . Hence k must be a perfect square. $\square$

$p(x)$ a polynomial with integer coefficients and such that both $p(0)$ and $p(1)$ are odd. Then $p(x)$ has no integer solutions.

(Unknown) Suppose a is an integer solution, then $a|p(0)$, hence a is odd. Now

$a-1|p(a)-p(1)=p(1)$, hence $a-1$ is odd, a contradiction.

$f(x)=(x-a)(x-b)(x-c)(x-d)-9$, where a, b, c, d are distinct integers. Suppose $f(x)$ has an integer solution. Show that $a+b+c+d$ is divisible by 4.

(Unknown) Suppose γ is an integer root, then $\gamma-a, \gamma-b, \gamma-c, \gamma-d$ are all distinct. But $0=(\gamma-a)(\gamma-b)(\gamma-c)(\gamma-d)-9$, hence $\gamma-a, \gamma-b, \gamma-c, \gamma-d$ can only be $\pm 1, \pm 3$ (no ± 9). Add then to get 0, i.e., $a+b+c+d$ divisible by 4.

$f(x)=a_0+a_1x+a_2x^2+\dots+a_nx^n$ and such that all its roots are real, also $a_i=\pm 1$.

Find all possible polynomials.

(Unknown) The roots are $x_1, x_2, \dots, x_n$. Then by Vieta,

$$x_1 + x_2 + \dots + x_n = -\frac{a_{n-1}}{a_n},$$

$$x_1x_2 + x_1x_3 + \dots + x_{n-1}x_n = \frac{a_{n-2}}{a_n}.$$

Hence

$$x_1^2 + x_2^2 + \dots + x_n^2 = (x_1 + x_2 + \dots + x_n)^2 - 2(x_1x_2 + x_1x_3 + \dots + x_{n-1}x_n)^2$$

$$= \left(-\frac{a_{n-1}}{a_n}\right)^2 - 2\frac{a_{n-2}}{a_n} = 1 - \frac{2a_{n-2}}{a_n}.$$

But $x_1^2 + x_2^2 + \dots + x_n^2 \geq n\sqrt{x_1^2x_2^2\dots x_n^2} = n$. Together we get $1 - \frac{2a_{n-2}}{a_n} \geq n$. Therefore

$$-1 \leq \frac{a_{n-2}}{a_n} \leq \frac{1-n}{2}, \text{ or } 3 \leq n. \text{ For } n=1, \text{ get } \pm(x \pm 1), \text{ for } n=2, \text{ get } \pm(x^2 \pm x - 1),$$

for $n=3$, get $\pm(x-1)^2(x+1), \pm(x-1)(x+1)^2$.

$p(x)$ an integer polynomial and let $n(p)$ the number of integers k such that $p(k)^2 = 1$. Show that $n(p) \leq \deg p + 2$.

(IMO1974) The statement is clear if the degree of $p(x) \leq 2$. Assume now $\deg p \geq 3$ and $n(p) > \deg p + 2$, i.e. $n(p) \geq \deg p + 3$. Hence there are at least 3 distinct integers a_1, a_2, a_3 such that $p(a_i) = 1$. Also there are at least 3 distinct integers b_1, b_2, b_3 such that $p(b_i) = -1$. Let $p(x) = (x-a_1)(x-a_2)(x-a_3)q(x) + 1$. Plug in b_1 etc. to get a contradiction.

System

Find all roots, real and complex, of the system

$$x + y + z = 3,$$

$$x^2 + y^2 + z^2 = 3,$$

$$x^3 + y^3 + z^3 = 3.$$

(USAMO1973) Regard them (x, y, z) as the roots of the equation $t^3 - at^2 + bt + c = 0$.

Then $a = x + y + z = 3$, $2b = 2(yz + zx + xy) = (x + y + z)^2 - x^2 - y^2 - z^2 = 6$. Put x, y, z into the equation and add them, get $3 - 3a + 3b - 3c = 0$, get $c = 1$, and the equation becomes $(t - 1)^3 = 0$.

Given

$$\begin{aligned}x + y + z &= 3, \\x^2 + y^2 + z^2 &= 5, \\x^3 + y^3 + z^3 &= 12.\end{aligned}$$

Find

$$x^4 + y^4 + z^4.$$

(unknown) 29.

Structures

Let $f(x)$ be a polynomial with real coefficients and such that $f(x) \geq 0$ for all real x . Show that $f(x)$ is the sum of squares of two real polynomials.

(Unknown) Let

$$f(x) = a_0(x - \alpha_1)^{e_1}(x - \alpha_2)^{e_2} \dots (x - \alpha_t)^{e_t}(x - \beta_1)(x - \beta_2) \dots (x - \beta_s)(x - \overline{\beta_1})(x - \overline{\beta_2}) \dots (x - \overline{\beta_s}),$$

where the α_i 's are real and the β_j 's are purely complex (nonreal) with conjugate

$\overline{\beta_j}$'s. Note that $a_0 > 0$, and the indices e_i 's are even, otherwise a small change

near the corresponding α_i 's will change the sign. Now $(x - \beta_1)(x - \beta_2) \dots (x - \beta_s) = u(x) + iv(x)$, with $u(x), v(x)$ real polynomials, and

$$(x - \overline{\beta_1})(x - \overline{\beta_2}) \dots (x - \overline{\beta_s}) = u(x) - iv(x). \text{ Thus}$$

$$(u(x) + iv(x))(u(x) - iv(x)) = u^2(x) + v^2(x). \text{ Also}$$

$$(x - \alpha_1)^{e_1}(x - \alpha_2)^{e_2} \dots (x - \alpha_t)^{e_t} = p^2(x) \text{ as mentioned. Thus}$$

$$f(x) = (\sqrt{a_0} p(x) u(x))^2 + (\sqrt{a_0} p(x) v(x))^2.$$

$5^{1985} - 1$ is the product of three factors, each larger than 5^{100} .

(Unknown) Note $x^5 - 1 = (x - 1)(x^4 + x^3 + x^2 + x + 1)$. Take $x = 5^{100}$, get

$$\begin{aligned}
x^4 + x^3 + x^2 + x + 1 &= x^4 + (6x^3 - 5x^3) + (11x^2 - 10x^2) + (6x - 5x) + 1 \\
&= (x^2 + 3x + 1)^2 - 5x(x+1)^2 \\
&= (x^2 + 3x + 1)^2 - 5^{199 \times 2} (x+1)^2 \\
&= (x^2 + 3x + 1 + 5^{199} (x+1))(x^2 + 3x + 1 - 5^{199} (x+1)),
\end{aligned}$$

we then get the three factors as required.

$x^4 + x^3 + x^2 + x + 1$ is the difference of the squares of two real polynomials (with different degrees) and there are four such solutions.

(Unknown) Let $x^4 + x^3 + x^2 + x + 1 = f^2(x) - g^2(x) = (f(x) + g(x))(f(x) - g(x))$. Clearly $\deg f > \deg g$, hence $\deg f = 2$ and $\deg g = 1$. Let $f(x) = x^2 + ax + b$, $g(x) = cx + d$, plug in and simplify, get $f(x) = x^2 + \frac{1}{2}x + 1$ and $g(x) = \frac{\sqrt{5}}{2}x$. (The four solutions correspond to $\pm f(x), \pm g(x)$).

$Q(t)$ a real polynomial with two distinct real roots. Show that there exists a non-constant polynomial with leading coefficient 1 such that the absolute values of all coefficients of $Q(P(x))$ other than the leading one are less than 0.001.

(IOM2nd2017) :Let $Q(t) = c((t-a)^2 - D)$, where a, c, D are constants, $c \neq 0$ and $D > 0$. It suffices to find a polynomial $R(x)$ with leading coefficient 1 and such that all coefficients of the polynomial $R(x)^2 - D$ except the leading one are less than $\delta = \frac{0.001}{|c|}$, then we may take $P(x) = R(x) + a$. We look for $R(x)$ of the form

$$R(x) = x^n + \varepsilon \sum_{i=1}^m (x^{n-a_i} + x^{a_i}) + b, \text{ where } m \text{ and } a_1 < a_2 < \dots < a_m < n \text{ are positive}$$

integers, b and ε are real numbers. The monomials that occur in the expansion of $R(x)^2$, other than x^{2n}, x^n and the constant term are

$x^{2n-a_i}, x^{2n-2a_i}, x^{2n-a_i-a_j}, x^{n+a_i}, x^{n-a_i+a_j}, x^{n-a_i}, x^{a_i+a_j}, x^{2a_i}$ and x^{a_i} . We set the numbers a_i

in such a way that each of these monomials occurs only once (or twice in $x^{a_i+a_j} = x^{a_j+a_i}$). If $n > 3a_m$, that is equivalent to choose numbers a_i so that the

numbers a_i and $a_j + a_k$ are pairwise different for all i, j, k , with $j \leq k$. We may

take $a_i = 3^{i-1}, 1 \leq i \leq m$ and $n = 3^m + 1$.

Note now the coefficient of $R(x)^2 - D$ at x^n is $2b + 2m\epsilon^2$. Thus each coefficient of $R(x)^2 - D$ is each of the numbers $0, 2\epsilon, \epsilon^2, 2b\epsilon, b^2 - D, 2b + 2m\epsilon^2$. We may take

$$b = -\sqrt{D}, \epsilon < \min\left\{\frac{\delta}{2b}, \frac{\delta}{2}, \sqrt{\frac{\delta}{2}}\right\} \text{ and } m = \left\lfloor \frac{-b}{\epsilon^2} \right\rfloor \text{ will do.}$$

Does there exists a polynomial $P(x)$ with integer coefficients and such that

$$P(1 + \sqrt[3]{2}) = 1 + \sqrt[3]{2} \text{ and } P(1 + \sqrt{5}) = 2 + 3\sqrt{5}?$$

(VIET2017) Let $Q(x) = P(1+x) - 1$. Then $Q(\sqrt[3]{2}) = \sqrt[3]{2}$ and $Q(\sqrt{5}) = 1 + 3\sqrt{5}$.

Therefore $Q(x) - x$ has an irrational root $\sqrt[3]{2}$, thus $Q(x) - x = (x^3 - 2)R(x)$. Now

$R(x)$ is a polynomial with integer coefficients, thus $R(x) = a + b\sqrt{5}$. Put into

$$Q(x), \text{ get } 1 + 3\sqrt{5} - \sqrt{5} = Q(\sqrt{5}) - \sqrt{5} = (5\sqrt{5} - 2)(a + b\sqrt{5}), \text{ or } 25b - 2a = 1,$$

$$2 = 5a - 2b, \text{ no integer solution.}$$

$r(x)$ a real polynomial of odd degree, there exist only finitely many pairs of real polynomials $p(x)$ and $q(x)$ such that $p(x)^3 + q(x^2) = r(x)$.

(1st IOM2016) Replace x by $-x$, get $p(x)^3 - p(-x)^3 = r(x) - r(-x) = u(x)$, check that $u(x)$ is odd, nonzero, has the same degree as $r(x)$, with $p(x) - p(-x)$ an odd divisor of $u(x)$. There are only finitely many divisors of $u(x)$ up to a constant factor. So it suffices to check that for any fixed odd divisor $xa_0(x^2)$ of $u(x)$ there

are only finitely many $p(x)$ such that $p(x) - p(-x)$ is proportional to $xa_0(x^2)$,

i.e., $p(x)$ is of the form $\lambda xa_0(x^2) + b(x^2)$Hard.

Find all polynomials $P(x, y)$ such that (i) P is homogeneous of degree n , i.e. $P(tx, ty) = t^n P(x, y)$, (ii) for all a, b, c , $P(a+b, c) + P(b+c, a) + P(c+a, b) = 0$, (i) $P(1, 0) = 1$.

(IMO1975) $P(x, y)$ is of the form $a_0 x^n + a_1 x^{n-1} y + \dots + a_{n-1} x y^{n-1} + a_n y^n$. For $n = 1$, get $P(x, y) = x - 2y$. Suppose $n > 1$. Put $x, 0, y$ into (ii), get

$P(x, y) + P(y, x) + P(x + y, 0) = 0$, hence $P(x, y) + P(y, x) = -(x + y)^n$. Put $x, x, -2x$ into (ii), get $P(2x, -2x) + P(-x, x) + P(-x, x) = 0$, or $(2^n - 2)P(x, -x) = 0$. As $n > 1$, get $P(x, -x) = 0$. Thus $P(x, y) = (x + y)P_1(x, y)$. As $P_1(x, y)$ and satisfies (i), (ii) and (iii), get $P_1(x, y) = (x + y)P_2(x, y)$. Proceed by induction to get $P(x, y) = (x + y)^{n-1}P_{n-1}(x, y)$. As $P_{n-1}(x, y)$ is of degree 1, get $P_{n-1}(x, y) = x - 2y$.
 Done.

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